v1.0Deep Dive_ One-Quark Universe Architecture

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Quark Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** `v1.0Deep Dive_ One-Quark Universe Architecture V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Top quark mass	m_t ~ 173 GeV	m_t = 172.52±0.33 GeV (LHC Run 3 2024). Metastability threshold at 171.7 GeV — universe is metastable	HIGH	Mass value critical for vacuum stability calculation — now more precise
Exotic quark states	qqq and qq̄ only	Tcc+ (2022), 3 pentaquarks (2023) confirmed by LHCb. EXOTIC_QUARK_CLUSTER with 4-5 quarks added	HIGH	V1.0 data model fundamentally incomplete for confirmed exotic states
QGP deconfinement	Not in scope	ALICE Run 3: T_c = 156 MeV lattice confirmed. V2.0: quark deconfinement above T_c — free color cloud visual	HIGH	Missing major QCD phase transition in simulation
String breaking probability	F(r) = sigma*r only	Lund model: string breaks with P = exp(-pi*m_q^2/sigma) per unit length per time — quark pair creation	MEDIUM	V1.0 lacks mechanism for string breaking and new particle pair creation

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Exotic, Top, QGP
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Deep Dive_ One-Quark Universe Architecture V1.0 archived | V2.0 is the active specification as of June 2026